

Suman Bhowmick

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Research Interest

My research interests lie in hydrology, groundwater dynamics, and hydro-climatic variability, with an emphasis on geospatial analysis and remote sensing. I focus on understanding rainfall patterns, land–water interactions, groundwater variability, and environmental processes using GIS and earth observation datasets. My work aims to contribute to data-driven approaches for analysing water resource dynamics and environmental change.

Education

Ph.D. (Ongoing) Presidency University, Kolkata CGPA: NA	2026–Ongoing
M.A./M.Sc in Geography North-Eastern Hill University, Shillong CGPA: 8.00/10	2022–2024
B.Sc in Geography Acharya Prafulla Chandra College, New Barrackpur CGPA: 8.46/10	2019–2022

Technical Skills

GIS	: ArcGIS Pro, ArcGIS, QGIS
Programming	: R, Python, LaTeX, Google Earth Engine
Data Analysis	: SPSS, Stata, NVivo, TerrSet
Image Processing	: Satellite Imagery, Aerial Photography

Projects & Web Applications

- **West Bengal Electoral Map Dashboard** *Aug 2025 – Aug 2026*
Developed a full-stack interactive GIS dashboard visualizing West Bengal's 294 constituencies. Integrated responsive Leaflet maps with smart data-matching algorithms to enrich live election results with historical candidate profiles.
Technologies: React.js, TypeScript, Leaflet.js, Tailwind CSS.
Link: [View Dashboard](#)
- **Personal Portfolio Website** *July 2025 – Present (Ongoing)*
Designed and developed a professional portfolio to showcase GIS projects, research, and technical skills. Implemented responsive design and interactive features for enhanced user experience.

Technologies: HTML, CSS, JavaScript, GitHub Pages.

Link: [View Portfolio](#)

Field Experience

Ri-Bhoi District, Meghalaya

Feb 2024 – Apr 2024

- Conducted field visits for systematic soil sampling across diverse topographic units.
- Used GPS devices for geotagging and elevation measurement of all sampling points and preparation of road map.
- Collected field notes on vegetation, barren and land use for each location to support GIS-based LULC modeling.

Research Engagement

Research Collaborator

Sep 2025 – Present

Meteorological Drought and Groundwater Anomaly Analysis using GRACE Data In collaboration with Mohan Sarkar (Ph.D. Scholar, IIT Bombay)

- Investigating groundwater variability and depletion patterns through integration of GRACE-derived datasets and climatic variables (precipitation and temperature).
- Applying time-series analysis and geospatial techniques to evaluate spatio-temporal drought trends.

Research Collaborator

Aug 2025 – Present

Integrated Machine Learning and AHP-based Drought Severity and Risk Mapping of Jharkhand (1991–2025) In collaboration with Animesh Pramanick (IIHS)

- Developed a hybrid ML–AHP framework for drought severity and risk assessment.
- Analysed long-term hydro-climatic datasets (1991–2025) to evaluate spatio-temporal drought variability.
- Generated drought severity and risk maps using GIS-based multi-criteria decision analysis.

Workshops and Training Programs

- **Advanced Image Analysis for Geospatial Professionals** [Online] Aug 2025
Coordinator: Dr. Poonam S. Tiwari (IIRS-ISRO) (1-day Workshop)
- **Transforming GIS with Python Integration** [Online] Jul 2025
Lecture by Krishna G. Lodha (GIS Enthusiast, Ronten Grass)
- **AI-ML for Earth Observation: Ground Segment Operations, Data Processing & Applications** [Online] Jun 2025
Bhoonidhi Workshop Series, NRSC, ISRO (1-day Workshop)
- **Deep Learning for SAR Image Processing** [Online] Jun 2025
Lecture by Dr. Florence Tupin (Télécom Paris),
IEEE GRSS and IEEE Madras Section

- **Road Map of Future Geographies: Drone Technology and Geo-Spatial Analysis**
Department of Geography, Adamas University (2-day Workshop)

Mar 2019
Kolkata, India

Online Courses

- **Hydraulic Modeling with HEC-RAS** Sep – Oct 2025
UNESCO Open Learning — USACE Open Hydrology Series
Introduced 2D hydraulic modeling using HEC-RAS, covering terrain data integration, mesh generation, boundary conditions, dam and levee breaching simulation, and result visualization techniques.
- **Remote Sensing and Numerical Ocean Modelling for Oceanic Processes** Aug 2025
IIRS-ISRO (5 Days, Coordinators: Dr. D. Mitra & Dr. Sachiko Mohanty)
Covered ocean circulation fundamentals, upwelling processes, wave modelling, remote sensing of oceanic processes, and ocean biogeochemistry.
- **Python for Data Science** Jul–Aug 2025
NPTEL (IIT Madras) — Instructor: Prof. Raghunathan Rengasamy
Equipped participants with Python programming skills for solving data science problems. Covered Spyder basics, Python data types, NumPy, Pandas, data visualization (Matplotlib, Seaborn), control structures, and a case study.
- **Writing in the Sciences** Jun–Jul 2025
Coursera (Stanford University) — Instructor: Prof. Kristin Sainani
Covered scientific writing structure, clarity, and publication ethics through hands-on assignments.
- **Remote Sensing Data Analytics for Crop Production Forecasting** Jun 2025
IIRS-ISRO (5 Days, Course No. 168, Coordinator: Dr. N. R. Patel)
Covered RS-based crop discrimination, yield estimation, and UAV-assisted crop simulation techniques.

Honors and Awards

- **UGC-NET Qualified** – Dec'24, June'25 and Dec'25
- **Top Rank Holder in M.Sc. Geography, NEHU** – 2024

Languages

- Bengali (Native), English (Fluent), Hindi (Fluent)